

AI Documentation

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Brainstorming for Proposal:

Asked AI: Generate 20 different research questions that use psychology and AI that incorporate a large public database. It can be about any topic of psychology but make sure it includes a knowledge mining method. Create a mix of broad and specific research questions.

AI Response:

Can **topic modeling (LDA)** apply to the Twitter Academic API dataset to predict shifts in collective anxiety during major global crises?

Can **multimodal deep learning** (text + image analysis) applied to publicly shared Instagram data identify behavioral precursors of eating disorders?

How can **graph neural networks** applied to the Human Connectome Project dataset classify personality dimensions from resting-state functional connectivity patterns?

Can **hierarchical Bayesian modeling combined with machine learning feature selection** on the World Values Survey dataset uncover latent psychological dimensions predicting authoritarian attitudes?

Using **dynamic topic modeling and temporal clustering** on the Google Books Ngram Viewer corpus, how have societal representations of mental illness evolved over the past century?

Using **semantic network analysis and transformer embeddings** on dialogue transcripts from the TalkLife public dataset, can AI identify conversational patterns predictive of crisis escalation?

Out of the 20 questions I asked ChatGPT to create, I have chosen my top 6 that had the most interesting topic to me, the best well-rounded question, or the most interesting dataset. From this search I was able to expand the types of datasets available to me and choose one that fits best with the research I want to conduct.

My brainstorming:

Twitter API dataset, can twitter message patterns identify suicidality before occurring?

Can the time tweets be sent correlate to a person's mood, and can we learn the fluctuations by tweet timings?

How can **gradient boosting models (e.g., XGBoost)** applied to structural MRI features predict working memory performance? Human Connectome Project, UK Biobank

Using longitudinal cognitive performance data from the UK Biobank, can **causal machine learning models (e.g., difference-in-differences combined with gradient boosting)** determine whether changes in attention-related task performance after 2020 are attributable to the COVID-19 pandemic itself, or better explained by mediating factors

such as stress, sleep disruption, screen time, or lack of social interaction, the rise of social medias like TikTok, or socioeconomic strain?

Using longitudinal cognitive performance and educational attainment data from the Programme for International Student Assessment (PISA) database, can **hierarchical Bayesian modeling combined with machine learning trend decomposition** determine whether observed declines in standardized cognitive scores reflect true reductions in general cognitive ability, or are better explained by changes in educational policy, socioeconomic inequality, digital media exposure, or test-taking behavior?

Using longitudinal literacy and demographic data from the National Assessment of Educational Progress (NAEP) database, can **multilevel gradient boosting models combined with causal mediation analysis** determine whether the recent increase in students reading below grade level is primarily attributable to pandemic-related learning disruption, increased digital media consumption, socioeconomic inequality, or shifts in instructional practices?

Theories Section: (4/27/26)

Write a 2-page paper about the theories used in my research about naep reading scores including social reproduction theory. Include 10 sources articles and explain their connection to the paper
Revised prompt: rework the article focusing on naep reading scores and how they relate to the drop in scores after the covid 19 pandemic

Put into chat gpt and gemini, pull the best sources from the prompt outputs into the theories section.

Ethics Section: (5/4/26)

Prompt: Ethical concerns for my research with NAEP reading scores and us census socioeconomic information

AI Output: Privacy and Confidentiality

Reidentify information

Small subgroup disclosure

Data linkage concerns

Ethical Practices

Deficit framing

Discuss policy implications

Equity and Social Justice

Ethical Risks

Ethical Practices

Statistical and Methodological Ethics

Overinterpretation

Ignoring sampling design

Using NAEP weights appropriately

Bias and Researcher Position

Theoretical frameworks

Cultural and socioeconomic differences

Data Representation

Policymaker misuse

Wording

Context

Compliance

IRB

Citation and transparency

Abstract (5/8/26)

Prompt: write an abstract 150 words. concluding that the findings suggest a broad post-2020 decline in reading scores across states, but not a perfectly uniform shock: in Grade 8, the post-2020 effect varies significantly with state median income. Across states, structural socioeconomic factors including poverty, racial composition, and educational attainment remain major correlates of reading achievement, alongside income and digital access. as well as all the other stuff that does into an abstract using the information provided to you.

Output (Chat GPT): The COVID-19 pandemic introduced unprecedented disruptions to education across the United States, raising concerns about long-term academic outcomes and

inequality. This study examines how state-level fourth- and eighth-grade reading scores, measured by the National Assessment of Educational Progress (NAEP), changed after 2020 and how post-pandemic outcomes relate to pre-existing socioeconomic conditions derived from the United States Census Bureau. Using post-2020 reading scores as the dependent variables and state socioeconomic indicators as independent variables, the analysis evaluates whether structural inequalities shaped pandemic learning outcomes. Findings suggest a broad national decline in reading achievement following 2020, though the impact was not uniform across states. In Grade 8, score changes vary significantly with state median income, indicating differentiated recovery patterns. Across states, poverty levels, racial composition, educational attainment, income, and digital access remain strong correlates of reading performance, highlighting the enduring influence of structural socioeconomic inequality on educational achievement.

One sentence thesis: Sensitive data refers to information related to individuals or groups that could result in privacy violations, stigma, discrimination, or social harm if improperly analyzed, interpreted, or disclosed.

Output (Gemini): This study investigates the shifting landscape of American literacy following the global disruptions of 2020, analyzing the intersection of socioeconomic variables and student performance. Utilizing state-level data, the research evaluates how traditional achievement gaps have evolved in the post-pandemic era. The findings suggest a broad post-2020 decline in reading scores across nearly all states; however, the data reveals that this was not a perfectly uniform shock. Notably, in Grade 8, the magnitude of the post-2020 effect varies significantly according to state median income, suggesting that wealthier states may have possessed greater resilience to instructional disruptions. Despite these recent shifts, structural socioeconomic

factors—including poverty rates, racial composition, and adult educational attainment—remain the primary correlates of reading achievement. Furthermore, the results highlight the growing importance of household income and digital access in facilitating literacy. These findings underscore the persistent influence of systemic inequality on educational outcomes in a changing national context.

I combined the best aspects of each while tailoring the abstract to our research more. I could have been more specific in the prompt but felt that the results sections of our paper did the heaviest lifting.

AI Assistance for Methods and Coding

All data collection and empirical analyses were implemented through scripted workflows to ensure transparency and reproducibility. NAEP reading scores were obtained via the NAEP Data API, and socio-economic covariates were collected from the American Community Survey (ACS) using the U.S. Census API. Data cleaning, variable construction (including rate measures and the post-2020 indicator), and panel dataset assembly were conducted in R using reproducible scripts.

The panel regression models—including pooled OLS, state fixed effects, and interaction specifications—were estimated using standard statistical packages with consistent model definitions across all specifications. All preprocessing steps, model inputs, and estimation procedures were fully scripted to enable replication of results from raw data sources.

Generative AI tools were used to support code documentation, improve clarity in the presentation of methodological descriptions, and organize analytical workflows.

Moreover, AI was utilized to help perform our key methods of our project including the XGBoost models, etc. AI Outputs were continuously improved through code reviews and cross checking with multiple AI platforms including Cursor, ChatGPT and Windsurf. Some major prompts include the following:

Implement an XGBoost model with the reading score being the target and the other variables to train on. Training should be on the pre-2020 years and testing should be on the post-2020 years. Do rolling validation on the training set to get the best model where pre-2020 year is tested on and all previous years before it are trained on. Let me know if this is best approach.

The grade binary has very high importance in our model, is it better if we made two models, one for grade 4 and one for grade 8 without the grade variable so we can see better importance of the main socioeconomic variables.

I want to create a new script for shap analysis to see how each variable contributes.

I want a outputs folder that includes all the overall and per variable metrics as well some plots like variable dependencies and overall shap. What am I missing here.

AI Assistance for Result, Discussion, and Conclusion Writing

The results, discussion, and conclusion sections of this paper were drafted with the assistance of Claude. The AI was used to structure and articulate empirical findings from the panel for regression analysis, generate academic prose from quantitative outputs, suggest relevant in-text citations, and refine section drafts for clarity. AI was mostly used for checking the grammar and formatting of the sections.

All statistical calculations, interpretations, and analytical decisions were verified and polished with the help of Claude. The AI served as an editing tool for the writing of these sections.